Mitoprep Premium Test 5

Date · 07 January 2026

Physics :- Units & Measurements, Motion in a Straight Line, Motion in a Plane,
Laws of Motion, Work Energy & Power

Chemistry :- Some Basic Concepts Of Chemistry, Structure of Atom,
Classification of Elements & Periodicity, Chemical Bonding

Biology :- The Living World, Biological Classification, Plant Kingdom,
Animal Kingdom, Structural Organisation in Animals

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

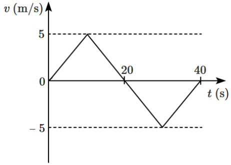
Physics

1. The dimensions $[MLT^{-2}A^{-2}]$ belong to the:
1. electric permittivity
2. magnetic flux
3. self-inductance
4. magnetic permeability

2. Rounding off the number 978.02 upto two significant figures will leave us with:
1. 98
2. 9.8×10^1
3. 9.8×10^2
4. 9.78×10^2

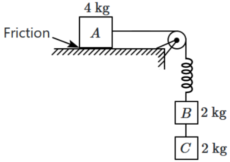
3. A particle is moving along the x-axis with its position (x) varying with time (t) as $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$. The ratio of its initial velocity to its initial acceleration is:
1. $2\alpha : \delta$
2. $\gamma : 2\delta$
3. $4\alpha : \beta$
4. $\gamma : 2\beta$

4. From the velocity-time $(v\text{-}t)$ plot shown in the figure, what is the distance travelled by the particle during the first 40 s?



1. zero
2. 100 m
3. 150 m
4. 200 m

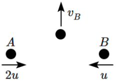
5. The system is at rest initially, due to the force of friction acting on A. If the string connecting the lower blocks is cut, the accelerations of the blocks A, B & C will be, respectively,



1.	$\frac{g}{3}$ to left, g upward, g downward.
2.	zero, zero, g downward.
3.	zero, g upward, g downward.
4.	g to right, zero, g downward.

6. A ball strikes against the floor and returns with double the velocity. In which type of collision is it possible?
1. Perfectly elastic
2. Inelastic
3. Perfectly inelastic
4. It is not possible

7. Two identical particles (A, B) moving in opposite directions with velocities $2u, u$ collide; and the final velocity of B is perpendicular to its initial direction. The velocity of A, along the original direction, is:



1.	u	2.	$\frac{u}{2}$
3.	$2u$	4.	$\sqrt{2}u$

8. A particle moves uniformly in a circle of radius R with a speed v. When it covers a semi-circle (for the first time), its average velocity is:

1.	v	2.	$\frac{v}{2}$
3.	$\frac{v}{\pi}$	4.	$\frac{2v}{\pi}$

9. A particle is constrained to move along the x-axis under the action of a force:

$\overrightarrow{F} = (2+3x)\hat{i}.$

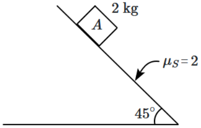
What is the work done by this force when the particle is displaced from $x=0$ to $x=4$ m?

1. 16 J
2. 32 J
3. 4 J
4. 8 J

10. Two particles are projected in the air at the same speed u at an angle θ_1 and θ_2 (θ_1 and θ_2 are acute angles) to the horizontal, respectively. If the maximum height reached by the first particle is greater than that of the second, then which of the following is correct?
(T_1 and T_2 are the times of flight of the two particles respectively)

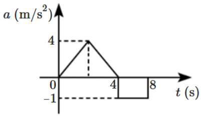
1. $\theta_1 > \theta_2$
2. $\theta_1 = \theta_2$
3. $T_1 < T_2$
4. $T_1 = T_2$

11. The coefficient of static friction between block A and the 45°-incline is, $\mu_S = 2$. The block A is released on the incline. The force of friction on the 2 kg block is: (take $g = 10$ m/s^2)



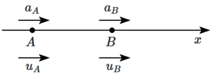
1. 40 N
2. $20\sqrt{2}$ N
3. $10\sqrt{2}$ N
4. 10 N

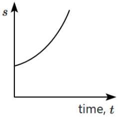
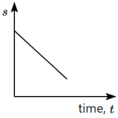
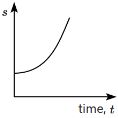
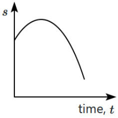
12. The acceleration-time graph of a particle is shown in the figure. What is the velocity of the particle at $t = 8$ s if the initial velocity of the particle is 3 m/s?



1. 4 m/s
2. 5 m/s
3. 6 m/s
4. 7 m/s

13. Two particles, A & B, move along the x-axis (as shown in the figure) with their initial velocities u_A, u_B and their accelerations a_A, a_B (constant accelerations). Their relative separation s is measured from A towards B, with the initial condition as depicted in the figure. Match the conditions mentioned in **Column-I** with the corresponding correct graph of s, in **Column-II**.



Column-I		Column-II	
(A)	$a_A = a_B \neq 0$	(I)	
(B)	$a_A > a_B, u_A < u_B$	(II)	
(C)	$a_A < a_B, u_A = u_B$	(III)	
(D)	$a_A < a_B, u_A < u_B$	(IV)	

1.	A-II, B-I, C-IV, D-III	2.	A-III, B-IV, C-II, D-I
3.	A-II, B-IV, C-III, D-I	4.	A-III, B-I, C-II, D-IV

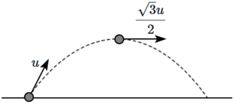
14. The motion of a particle is given by $S = 1 + 4t - 2t^2$. The distance travelled by the particle during $t = 0$ to $t = 2$ seconds will be:

1.	0 unit	2.	2 unit
3.	4 unit	4.	3 unit

15. A man (A) has to throw a ball vertically up to a partner (B) who is standing up, above his level by 15 m. The (B) partner can catch the ball only when it comes downwards with a maximum speed of $10\ \mathrm{m/s}$
(take acceleration due to gravity as $10\ \mathrm{m/s^2}$)
The minimum and maximum speeds of the throw are: (nearly)
1. $10\ \mathrm{m/s}$ and $20\ \mathrm{m/s}$
2. $10\ \mathrm{m/s}$ and $30\ \mathrm{m/s}$
3. $20\ \mathrm{m/s}$ and $20\sqrt{3}\ \mathrm{m/s}$
4. $10\sqrt{3}\ \mathrm{m/s}$ and $20\ \mathrm{m/s}$

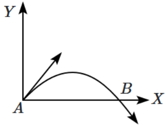
16. The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assume that bullet is facing constant resistance during its motion in the block. The bullet stops completely after travelling at $(4 + x)$ cm inside the block. The value of x is:
1. 2.0
2. 1.0
3. 0.5
4. 1.5

17. A projectile is launched from a horizontal surface with an initial velocity u. At its maximum height, the velocity of the projectile is $\frac{\sqrt{3}u}{2}$. The time of flight of the projectile is:



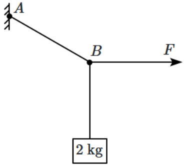
1.	$\frac{\sqrt{3}u}{g}$	2.	$\frac{2u}{g}$
3.	$\frac{u}{g}$	4.	$\frac{u}{2g}$

18. The velocity of a projectile at the initial point A is $2\hat{\imath}+3\hat{\jmath}$ m/s. Its velocity (in m/s) at the point B is:



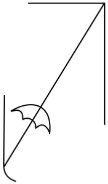
1.	$-2\hat{\imath}+3\hat{\jmath}$	2.	$2\hat{\imath}-3\hat{\jmath}$
3.	$2\hat{\imath}+3\hat{\jmath}$	4.	$-2\hat{\imath}-3\hat{\jmath}$

19. A 2 kg block is attached to a string, which is connected to a point A. A horizontal force F is applied to the string at B, so that AB makes an angle of 30° with the vertical, in equilibrium. The force F equals: $(g=10\ \mathrm{m/s^2})$



1.	10 N	2.	$10\sqrt{3}$ N
3.	$\dfrac{20}{\sqrt{3}}$ N	4.	$20\sqrt{3}$ N

20. Rain is falling vertically downward with a speed of $35\ \mathrm{m/s}$. The wind starts blowing after some time with a speed of $12\ \mathrm{m/s}$ in the east to the west direction. The direction in which a boy standing at the place should hold his umbrella is:



1.	$\tan^{-1}\left(\frac{12}{37}\right)$ with respect to rain
2.	$\tan^{-1}\left(\frac{12}{37}\right)$ with respect to wind
3.	$\tan^{-1}\left(\frac{12}{35}\right)$ with respect to rain
4.	$\tan^{-1}\left(\frac{12}{35}\right)$ with respect to wind

21. Given below are two statements:

Assertion (A):	It is difficult to move a bicycle along a road with its brakes on.
Reason (R):	When a bicycle moves with its brakes on, it skids and sliding friction is greater than rolling friction.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

22. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk upon the moving escalator will be:

1.	$\frac{t_1 t_2}{t_2 - t_1}$	2.	$\frac{t_1 t_2}{t_2 + t_1}$
3.	$t_1 - t_2$	4.	$\frac{t_1 + t_2}{2}$

23. Planck's constant (h), speed of light in the vacuum (c), and Newton's gravitational constant (G) are the three fundamental constants. Which of the following combinations of these has the dimension of length?

1.	$\frac{\sqrt{hG}}{c^{3/2}}$	2.	$\frac{\sqrt{hG}}{c^{5/2}}$
3.	$\frac{\sqrt{hG}}{G}$	4.	$\frac{\sqrt{Gc}}{h^{3/2}}$

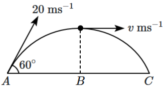
24. A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be:
1. 1.4%
2. 1.2%
3. 1.3%
4. 1.6%

25. The value of Stefan's constant (σ) is determined by experimentally measuring the remaining quantities in the equation: $E = A\sigma T^4$.
Both E, A are measured to an accuracy of 1%, while temperature T is measured to 0.5%. The error in the determination of σ is:
1. 2%
2. 4%
3. 2.5%
4. 0.5%

26. A block of mass m is placed over a rotating platform at a distance of R from the centre. The platform is rotating with constant angular velocity ω about its main axis. If the coefficient of friction between block and the platform is μ, then maximum value of R so that block will not slip over the platform will be:

1.	$\frac{\mu g}{\omega^2}$	2.	$\frac{\mu g^2}{\omega}$
3.	$\sqrt{\frac{\mu g^2}{\omega}}$	4.	$\sqrt{\frac{\mu g}{\omega^2}}$

27. A ball is projected from point A with velocity 20 ms^{-1} at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity v (in ms^{-1}) of the ball will be:

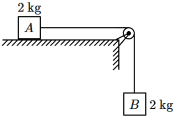


1.	20	2.	$10\sqrt{3}$
3.	zero	4.	10

28. What is the unit of the quantity represented by: $\frac{\text{(Angular momentum)}}{\text{(electric charge)}^2}$
1. Ω (ohm)
2. s/m
3. H (henry)
4. F (farad)

29. A lift of mass 500 kg starts moving downwards with an initial speed of 2 m/s and accelerates at a rate of $2\ m/s^2$. The kinetic energy of the lift when it has moved 6 m down, is:
1. 3 J
2. 3 kJ
3. 7 J
4. 7 kJ

30. The 2 kg block (A) lies on a smooth horizontal table and is connected by a light string passing over a smooth light pulley. The string is connected to a second 2 kg block (B). The system is released from rest. Take $g = 10\ \mathrm{m/s^2}$.



The tension in the connecting string is:

1.	zero
2.	20 N
3.	less than 20 N (non-zero)
4.	greater than 20 N

31. A ball, thrown vertically upward, is observed to move upward with a speed v_1 at time t_1 and a speed v_2, downward, at time t_2.
The average velocity of the ball during the motion is (downward):

1.	$\frac{v_2-v_1}{2}$	2.	$\frac{v_2+v_1}{2}$
3.	$v_2 - v_1$	4.	$v_2 + v_1$

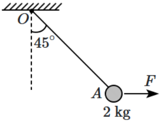
32. Which of the following motions of an object always results in zero acceleration?

1.	Any motion in a straight line.
2.	Projectile motion.
3.	Any motion in a circle.
4.	None of the motions above guarantees zero acceleration.

33. A boat is rowed across a river so that its velocity with respect to water is directed perpendicular to the river's flow. The speed of the boat with respect to the water is $2\ \mathrm{km/h}$ and the speed of the river's flow is $3\ \mathrm{km/h}$. If the river is 1 km wide, how far down the river will the passengers have to disembark, with respect to their starting point?

1. $\frac{2}{3}$ km
2. 1 km
3. $\frac{3}{2}$ km
4. None of the above is correct.

34. A simple pendulum consists of a bob (A) of mass 2 kg connected by means of a light, inextensible string of length 1 m to a fixed point O on the ceiling. A horizontal force is applied to the bob (A); the string makes an angle of 45° with the vertical & the system is stationary. Take $g = 10\ \mathrm{m/s^2}$, if required.



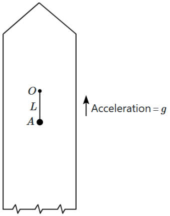
If the string is suddenly cut, the acceleration of the bob will be:

1. $10\ \mathrm{m/s^2}$
2. $20\ \mathrm{m/s^2}$
3. $10\sqrt{2}\ \mathrm{m/s^2}$
4. $20\sqrt{2}\ \mathrm{m/s^2}$

35. A car is negotiating a curved road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyre of the car and the road is μ_s. The maximum safe velocity on this road is:

1. $\sqrt{gR\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$
2. $\sqrt{\frac{g}{R}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$
3. $\sqrt{\frac{g}{R^2}\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$
4. $\sqrt{gR^2\left(\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}\right)}$

36. A simple pendulum of length L is suspended from a point O in a rocket which is ready to be launched from earth. The rocket takes off with an upward acceleration equal to g. Immediately after take off, the bob of the pendulum is given a horizontal velocity so that it just completes a vertical circle. The bob's speed (relative to the rocket) at its highest point is v_H, where:



1. $v_H = \sqrt{gL}$
2. $v_H = \sqrt{2gL}$
3. $v_H = \sqrt{3gL}$
4. $v_H = \sqrt{5gL}$

37. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?
1. $\sqrt{2gR}$
2. $\sqrt{3gR}$
3. $\sqrt{5gR}$
4. $\sqrt{gR}$

38. The quantities which have the same dimensions as those of solid angle are:
1. stress and angle
2. strain and arc
3. angular speed and stress
4. strain and angle

39. Consider the following statements about friction:

I.	Static friction can speed an object up from rest.
II.	Kinetic friction can prevent an object from slipping on a surface.
III.	Kinetic friction can slow an object down to rest.

Which of the following options correctly identifies the true statement(s)?

1.	**I** and **III** only	2.	**III** only
3.	**II** and **III** only	4.	**I, II,** and **III**

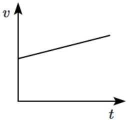
40. The potential energy of a particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$ where A and B are positive constants and r is the distance of the particle from the centre of the field. For stable equilibrium, the distance of the particle is:

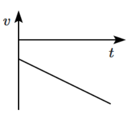
1.	$\frac{B}{A}$	2.	$\frac{B}{2A}$
3.	$\frac{2A}{B}$	4.	$\frac{A}{B}$

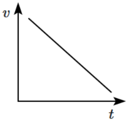
41. Given that T stands for time period and l stands for the length of simple pendulum. If g is the acceleration due to gravity, then which of the following statements about the relation $T^2 = l/g$ is correct?

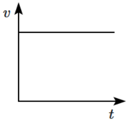
1.	It is correct both dimensionally as well as numerically.
2.	It is neither dimensionally correct nor numerically.
3.	It is dimensionally correct but not numerically.
4.	It is numerically correct but not dimensionally.

42. A car moving in a positive x-direction with uniform positive acceleration. Which one of the following velocity-time graphs correctly represents the motion of the car?

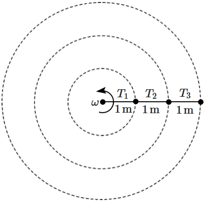
1. 

2. 

3. 

4. 

43. Three balls, each of mass 1 kg, are attached to three light strings, each of length 1 m, as shown in the figure. They are rotated in a horizontal circle with an angular velocity $\omega = 4$ rad/s. Which of the following options is correct?



1.	$T_1 = 16$ N, $T_2 = 32$ N, $T_3 = 48$ N
2.	$T_1 = 48$ N, $T_2 = 32$ N, $T_3 = 16$ N
3.	$T_1 = 96$ N, $T_2 = 80$ N, $T_3 = 48$ N
4.	$T_1 = 16$ N, $T_2 = 80$ N, $T_3 = 48$ N

44. The velocity v of a particle at a time t is given by $v = at + \dfrac{b}{t+c}$. The dimensions of a, b and c are respectively:

1. $\left[LT^{-2}\right], [L], [T]$
2. $\left[L^2\right], [T], \left[LT^2\right]$
3. $\left[LT^2\right], [LT], [L]$
4. $[L], [LT], \left[T^2\right]$

45. Consider a particle moving along a straight line under the action of a force which delivers constant power.

Assertion (A):	If the velocity of the particle is doubled, its acceleration is halved.
Reason (R):	The power delivered by the force equals the force $\times$ velocity which is a constant, and the acceleration is proportional to the force.

1.	**(A)** is True but **(R)** is False.
2.	**(A)** is False but **(R)** is True.
3.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
4.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.

Chemistry

46. Consider the given two statements:

Assertion (A):	The spectrum of He^+ ion is expected to be similar to that of hydrogen.
Reason (R):	He^+ is also a one-electron system.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A).**
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A).**
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

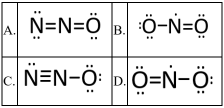
47. 4.74 g of an inorganic compound contains 0.39g of K, 0.27 g of Al, 1.92 g of SO_4 radicals and 2.16 g of water. If molar mass of the compound is 948 g mol^{-1}, the molecular formula of the inorganic compound is:

1.	$KAl(SO_4)_2 \cdot 12H_2O$
2.	$K_2Al_2(SO_4)_6 \cdot 12H_2O$
3.	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
4.	$K_2SO_6 \cdot Al_2(SO_4)_3 \cdot 12H_2O$

48. Which one of the following statements is incorrect related to Molecular Orbital Theory?

1.	The π^* antibonding molecular orbital has a node between the nuclei.
2.	In the formation of a bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.
3.	Molecular orbitals obtained from $2P_x$ and $2P_y$ orbitals are symmetrical around the bond axis.
4.	A $\pi-$bonding molecular orbital has larger electron density above and below the internuclear axis.

49. The following resonating structures for nitrogen(I) oxide are suggested:



Correct resonating structure(s) is/are:
1. A and C
2. Only A
3. A and B
4. C and D

50. Consider the given two statements:

Statement I:	The law of multiple proportions is followed by the gases SO and SO , but not by CO and CO_2.
Statement II:	Dalton's theory explains the laws of chemical combination as well as laws of gaseous volumes.

Select the correct option:

1.	**Statement I** is correct, and **Statement II** is correct.
2.	**Statement I** is incorrect, and **Statement II** is correct.
3.	**Statement I** is correct, and **Statement II** is incorrect.
4.	**Statement I** is incorrect, and **Statement II** is incorrect.

51. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :
(Given atomic mass of A = 64; B = 40; C = 32u)
1. ABC_3
2. AB_2C_2
3. ABC_4
4. A_2BC_2

52.

Assertion (A):	Na^+ and Al^{3+} are isoelectronic but the magnitude of ionic radius of Al^{3+} is less than that of Na^+.
Reason (R):	The magnitude of effective nuclear charge of the outer shell electrons in Al^{3+} is greater than that in Na^+.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True and **(R)** is False.
4.	Both **(A)** and **(R)** are False.

53. Maximum number of electrons in a subshell with l = 3 and n = 4 is:
1. 14
2. 16
3. 10
4. 12

54. How much oxygen gas (O_2) at standard temperature (0°C) and pressure (1 atm) is required to completely combust 1 liter of propane gas (C_3H_8) under the same conditions?

1.	7L	2.	6L
3.	5L	4.	10L

55. Which pair of ions from the following list do not constitute an iso-electronic pair?
1. Mn^{2+}, Fe^{3+}
2. Fe^{2+}, Mn^{2+}
3. O^{2-}, F^-
4. Na^+, Mg^{2+}

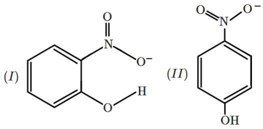
56. A 1 M solution of a compound 'X' has a density of 1.25 g/mL. If the molar mass of compound X is 85 g, what is the molality (m) of the solution?

1.	0.705 m	2.	1.208 m
3.	1.165 m	4.	0.858 m

57. Choose the option containing all isoelectronic species:
1. N^{3-}, O^{2-}, F, Na
2. $\mathrm{S^{-2}, Cl^-, K^+, Ca^{2+}}$
3. $\mathrm{NH_3, CH_4, PF_5, Na^+}$
4. $\mathrm{Ne, Na^+, F, N^{3-}}$

58. Which properties, among the four given below, of halogens increase as we move from iodine to fluorine?
(a) Bond length
(b) Electronegativity
(c) The ionization energy of the element
(d) Oxidizing power
1. **a**, **b** and **c**
2. **b**, **c** and **d**
3. **c**, **d** and **a**
4. **a** and **b** only

59. What type of hydrogen bonding exists in the given structures (I) and (II), respectively?



1.	Intermolecular H-boding & Intramolecular H-bonding
2.	Intramolecular H-boding & Intermolecular H-bonding
3.	Both are Intermolecular H-bonding
4.	Both are Intramolecular H-bonding

60. Consider the following reaction:
$\mathrm{4HNO_3(l) + 3KCl(s) \rightarrow}$
$\mathrm{Cl_2(g) + NOCl(g) + 2H_2O(g) + 3KNO_3(s)}$
How much $\mathrm{HNO_3}$ (in grams) is required to produce 110.0 g of $\mathrm{KNO_3}$?
(Given: Atomic masses of H, O, N, and K are 1, 16, 14, and 39, respectively.)
1. 32.2g
2. 69.4g
3. 91.5g
4. 162.5g

61. Given below are two statements:

Assertion (A):	Electron gain enthalpy of oxygen is more than that of sulphur.
Reason (R):	Oxygen is more electronegative than sulphur.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is False but **(R)** is True.
4.	Both **(A)** and **(R)** are False.

62. Factor among the following that does not affect the ionization enthalpy of an element is:
1. Effective nuclear charge
2. Atomic radius
3. Half-filled and completely filled sub-shell
4. Atomic number

63. The mass of H_2O formed by the reaction of 11.2 L H_2 and excess O_2 at STP will be:
1. 18 g
2. 9 g
3. 36 g
4. 4.5 g

64. A golf ball has a mass of 40g, and a speed of 45 m/s. If the speed can be measured within the accuracy of 2%, the uncertainty in the position is:
1. 14.6×10^{-23} m
2. 14.6×10^{-33} m
3. 1.46×10^{-23} m
4. 1.46×10^{-33} m

65. The number of d -electrons in Fe^{2+} (atomic number Z = 26) is different from the number of:
1. s-electrons in Mg (Z = 12)
2. p-electrons in Cl (Z = 17)
3. d-electrons in Fe (Z = 26)
4. p-electrons in Ne (Z = 10)

66. The mole ratio of H_2 and O_2 gas is 8:1. The ratio of their weight will be:
1. 1 : 1
2. 2 : 1
3. 4 : 1
4. 1 : 2

67. The mass of ammonia in grams produced when 2.8 kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen is:

1.	3600 g	2.	3000 g
3.	3400 g	4.	4000 g

68. Consider the following statements:

I:	The radius of an anion is larger than that of the parent atom.
II:	The electronegativity of any given element is a constant, and it is a measurable quantity.
III:	The electronegativity of an element is the tendency of an isolated atom to attract an electron.

Which of the above statements is/are correct?
1. Only **I**
2. Only **II**
3. Only **I** and **III**
4. Only **II** and **III**

69. In Heisenberg's uncertainty principle, if uncertainty in the position (Δx) is equal to the uncertainty in the momentum (Δp) then uncertainty in the velocity (Δv) will be:

1.	$\frac{1}{m}\sqrt{\frac{h}{2\pi}}$	2.	$\frac{1}{2m}\sqrt{\frac{h}{\pi}}$
3.	$\frac{1}{m}\sqrt{\frac{h}{\pi}}$	4.	$\frac{1}{2m}\sqrt{\frac{h}{2\pi}}$

70. The hybridization of XeF_4, SF_4 and $BrCl_3$, respectively, among the following is:
1. sp^3, sp^3, sp^3
2. dsp^2, sp^3, sp^3
3. sp^3d^2, sp^3d, sp^3d
4. d^2sp^2, sp^3d, sp^3d

71. According to Bohr's theory, what is the angular momentum of an electron located in the fifth orbit?
1. $25\frac{h}{\pi}$
2. $1.0\frac{h}{\pi}$
3. $10\frac{h}{\pi}$
4. $2.5\frac{h}{\pi}$

72. When 2 g of magnesium reacts with an excess of HCl, then H_2 gas and $MgCl_2$ are produced.
The volume of H_2 gas produced is X × 10^{-2} litres at STP.
The value of X is: (Nearest Integer)

1.	195	2.	186
3.	182	4.	179

73.

Assertion (A):	Ionic character in NaCl is less than that in CsCl.
Reason (R):	NaCl has more lattice energy than that in CsCl.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

74. The shortest wavelength of the Paschen series for H-atom is:

1.	$\frac{9}{R}$	2.	$\frac{16}{R}$
3.	$\frac{144}{7R}$	4.	$\frac{7R}{144}$

75. Which of the following compounds has the smallest bond angle?
1. SO_2
2. H_2O
3. H_2S
4. NH_3

76. Which of the following atoms are not accurately described by Bohr's model?
(a) H-atom
(b) D-atom
(c) T-atom
(d) He-atom
(e) Li- atom
Choose the correct option:

1.	**(a)** and **(b)**	2.	**(b)** and **(c)**
3.	**(a)**, **(b)** and **(c)**	4.	**(d)** and **(e)**

77. Which one of the elements has highest ionisation energy?
1. [Ne] $3s^2\ 3p^1$
2. [Ne] $3s^2\ 3p^2$
3. [Ne] $3s^2\ 3p^3$
4. [Ar] $3d^{10}\ 4s^2\ 4p^2$

78. What is the energy (in kJ/mol) of one mole of electrons when exposed to radiation with a frequency of 2×10^{14} Hz?
1. 436.1
2. 89.8
3. 79.8
4. 624.5

79. Which of the following law illustrates hydrogen and oxygen combination to form H_2O_2 and H_2O containing 5.93% and 11.2% hydrogen, respectively?
1. Law of conversation of mass
2. Law of constant proportion
3. Law of reciprocal proportion
4. Law of multiple proportions

80. A molecule, among the following, that obeys the octet rule is:

1.	PF_5	2.	SF_6
3.	CCl_4	4.	BH_3

81. The incorrect electronic configuration among the following is/are :
A. K $= [\mathrm{Ar}]4s^1$
B. Pd $= [\mathrm{Kr}]4d^8, 5s^2$
C. Cr $= [\mathrm{Ar}]3d^4, 4s^1$
D. Cu $= [\mathrm{Ar}]3d^{10}, 4s^1$
1. C and D only
2. B and C only
3. A and D only
4. B, C, and D only

82. The correct increasing order of B, Al, Mg, and K based on their metallic character is:
1. B > Al > Mg > K
2. Al > Mg > B > K
3. Mg > Al > K > B
4. K > Mg > Al > B

83. Match the coordination number and type of hybridization with the distribution of hybrid orbitals in space based on Valence bond theory.

	Coordination number and type of hybridisation		Distribution of hybrid orbitals in space
(a)	4, sp^3	(i)	Trigonal bipyramidal
(b)	4, dsp^2	(ii)	Octahedral
(c)	5, sp^3d	(iii)	Tetrahedral
(d)	6, d^2sp^3	(iv)	Square planar

	(a)	(b)	(c)	(d)
1.	(ii)	(iii)	(iv)	(i)
2.	(iii)	(iv)	(i)	(ii)
3.	(iv)	(i)	(ii)	(iii)
4.	(iii)	(i)	(iv)	(ii)

84. An isostructural compound with XeF_2 among the following is:
1. ICl_2^-
2. $SbCl_3$
3. $BaCl_2$
4. TeF_2

85. If the shortest wavelength in the Lyman series of a hydrogen atom is A, then what is the longest wavelength in the Paschen series of He^+?

1.	$\frac{5\ A}{9}$	2.	$\frac{9\ A}{5}$
3.	$\frac{36\ A}{5}$	4.	$\frac{36\ A}{7}$

86. Which of the following sets does **not** contain isoelectronic species?

1.	$BO_3^{3-}, CO_3^{2-}, NO_3^-$	2.	$SO_3^{2-}, CO_3^{2-}, NO_3^-$
3.	$(CN)^-, N_2, C_2^{2-}$	4.	$PO_4^{3-}, SO_4^{2-}, ClO_4^-$

87. The total number of neutrons present in 7 mg of ^{14}C is:
1. 24.09×10^{22}
2. 24.09×10^{20}
3. 2.409×10^{23}
4. 2.409×10^{19}

88. Ejection of the photoelectron from a metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is:
1. 4.0 eV
2. 5.5 eV
3. 4.5 eV
4. 5.0 eV

89. Beryllium has higher ionization enthalpy than boron. This can be explained as:

1.	Beryllium has a higher size than boron, hence its ionization enthalpy is higher.
2.	Penetration of 2p-electrons to the nucleus is more than the 2s electrons.
3.	It is easier to remove electrons from 2p-orbital as compared to 2s-orbital due to more penetration of s electron.
4.	Ionization energy increases in a period.

90. Which species among the following exhibits the highest electron affinity?
1. F^-
2. O^-
3. O
4. Na

Biology

91. The kingdom Protista includes a variety of organisms, among which the following is true about the group Chrysophytes?
1. They have a silica shell
2. They lack a nucleus
3. They are predominantly parasitic
4. They reproduce only asexually

92. What feature of Rhodophyceae enables them to survive at greater depths in the ocean?

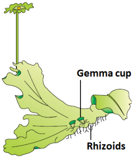
1.	Storage of floridean starch for high-pressure environments
2.	Presence of the pigment r-phycoerythrin, which absorbs light in the blue-green spectrum
3.	Ability to perform anaerobic respiration in low-oxygen conditions
4.	Reproduction by non-motile spores adapted for deep-water currents

93.

Statement I:	Red algae do not occur in well-lighted regions close to the surface of water.
Statement II:	Red algae reproduce by non-motile spores and non-motile gametes.

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is correct; **Statement II** is incorrect
3.	**Statement I** is incorrect; **Statement II** is correct
4.	**Statement I** is incorrect; **Statement II** is incorrect

94. The figure shows:



1. Female thallus of *Marchantia*
2. Male thallus of *Marchantia*
3. Gametophyte of *Funaria*
4. Sporophyte of *Funaria*

95. Which of the following is not correct regarding differences between liverworts and mosses in general?

		Liverworts	Mosses
1.	Rhizoids	Unicellular	Multi-cellular
2.	Sporophyte	Less elaborate	More elaborate
3.	Asexual reproduction	Gemmae	Budding in secondary protonema
4.	Dominant generation	Gametophyte	Sporophyte

96. What would be true for Ctenophora?

I:	Exclusively marine animals.
II:	Many members exhibit bioluminescence.
III:	Animals are diploblastic
IV:	Only sexual reproduction seen.

1. Only **I**, **II** and **III**
2. Only **I**, **III** and **IV**
3. Only **II**, **III** and **IV**
4. **I**, **II**, **III** and **IV**

97. Statocysts, present in some animals, help directly in:
1. Locomotion
2. Respiration
3. Feeding
4. Balancing

98. All the following are examples of mosses except:

1.	*Funaria*	2.	*Polytrichum*
3.	*Marchantia*	4.	*Sphagnum*

99. Which of the following is a correct match between the biological name and the order of the organism?
1. *Homo sapiens* – Hominidae
2. *Mangifera indica* - Anacardiaceae
3. *Triticum aestivum* - Poaceae
4. *Mangifera indica* - Sapindales

100. Dicotyledonae represents
a. Class
b. Order
c. Division
d. Genus

101. You discover an organism and observe that it shows the presence of the following characteristics:

A:	Bilateral symmetry
B:	Body covered with a plate like exoskeleton
C:	Legs with joints and wings

You conclude that this must be an arthropod but want to get more confident by observing another feature. Which of the following, if present, will strengthen your argument?

1.	Closed circulation
2.	Lack of a true coelom
3.	Ammonotelic excretion
4.	Body divided into head, thorax and abdomen segments

102. Which one of the following is true for fungi?
1. They lack a rigid cell wall
2. They are heterotrophs
3. They lack nuclear membrane
4. They are phagotrophs

103. Regarding pteridophytes:

I:	They are the first terrestrial plants to possess true vascular tissue
II:	Main plant is sporophyte
III:	Water is required for transfer of antherozoids

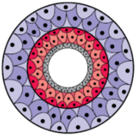
1. Only **II** is incorrect
2. Only **I** is correct
3. **I** and **II** are incorrect
4. **I**, **II** and **III** are correct

104. What would be true [T] or false [F] regarding amphibians?

I:	Skin is moist and without scales.
II:	Eyes have eyelids.
III:	A tympanum represents the ear.
IV:	Respiration is by gills, lungs and through skin.

	I	**II**	**III**	**IV**
1.	T	T	F	F
2.	F	F	F	T
3.	T	F	T	T
4.	T	T	T	T

105. The figure shows cross-section of the body of an animal. This animal:



I:	lacks a coelom
II:	is triploblastic
III:	is expected to have bilateral symmetry
IV:	is expected to be dorso-ventrally flattened

1. Only **I**, **II** and **III** are correct
2. Only **I**, **III** and **IV** are correct
3. Only **II**, **III** and **IV** are correct
4. **I**, **II**, **III** and **IV** are correct

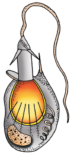
106. Nomenclature is
a. assigning a single name to an organism valid all over the world.
b. Defining properties of an organism
c. Localizing the area of abundance for that organism
d. Assigning three sets of name applicable at different levels of classification.

107. In which group of organisms the Cell walls form two thin overlapping shells which fit together?

1.	Chrysophytes	2.	Euglenoids
3.	Dinoflagellates	4.	Slime moulds

108. Biodiversity means-
a. Number of organisms known
b. Types of organisms known
c. Number of species known
d. All are correct

109. The structure seen in the given figure is present in:



1. Sponges
2. Flatworms
3. Cnidarians
4. Ctenophore

110. Which statement correctly differentiates between Urochordates and Cephalochordates?

1.	Urochordates retain their notochord throughout their life, whereas Cephalochordates lose their notochord during the larval stage.
2.	Cephalochordates, such as lancelets, retain a notochord extending the full length of their body into adulthood, whereas Urochordates, like tunicates, only possess a notochord during their larval stage.
3.	Both Urochordates and Cephalochordates exhibit a complex nervous system comparable to that of higher vertebrates.
4.	Urochordates are primarily terrestrial, whereas Cephalochordates are exclusively aquatic.

111. Which animals are diploblastic and show radial symmetry but lack cnidoblasts?
1. Cnidarians
2. Ctenophores
3. Echinoderms
4. Hemichordates

112. Consider the given two statements:

Statement I:	Cephalochordates retain their notochord throughout life.
Statement II:	They exhibit a dorsal tubular nerve cord and pharyngeal slits.

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is incorrect; **Statement II** is incorrect
3.	**Statement I** is correct; **Statement II** is incorrect
4.	**Statement I** is incorrect; **Statement II** is correct

113. Consider the following statements:

I.	Higher the taxa, more are the characteristics that members whithin the taxon share.
II.	Lower the category, greater is the difficulty of determining the relationship to other taxa at the same level.

Of the two statements:

1.	**I** is true but **II** is false
2.	**I** is false but **II** is true
3.	Both **I** and **II** are true
4.	Both **I** and **II** are false

114. Match each item in Column I with one in Column II regarding *Cycas* and select the correct match from the codes given:

	Column I [Feature of *Cycas*]		**Column II [Description]**
A	Root symbiont	P	Mycorrhiza
B	Leaves	Q	Cyanobacteria
C	Stems	R	Palmate, fall early
		S	Pinnate, persist for few years
		T	Unbranched
		U	Branched

Codes:

	A	**B**	**C**
1.	Q	S	T
2.	Q	S	U
3.	Q	R	T
4.	P	S	U

115. Identify the correct statements regarding the class Osteichthyes:

A.	Osteichthyes include both marine and freshwater fishes with a bony endoskeleton.
B.	Members of this class have a two-chambered heart and their skin is covered with cycloid or ctenoid scales.
C.	Osteichthyes are cold-blooded animals, have an air bladder to regulate buoyancy, and their fertilization is usually internal.

1.	**Statement A** and **Statement B** are correct
2.	**Statement B** and **Statement C** are correct
3.	**Statement A** and **Statement C** are correct
4.	**Statement A**, **B** and **C** are correct

116. In plant life cycle, the gametophyte is the generation that:
1. is diploid in genetic constitution
2. produces the gametes
3. produces the spores
4. has vascular tissue

117. Which one of the following pair of animals comprises 'jawless fishes'?

1.	Lampreys and eels
2.	Mackerals and rohu
3.	Lampreys and hagfishes
4.	Guppies and hagfishes

118. The suffix that denotes the taxonomic category 'Family' in plants would be
(1) -ae
(2) -aceae
(3) -ales
(4) -nae

119. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I		**Column II**
A	*Ulothrix*	P	Brown algae with diplontic life cycle
B	*Ectocarpus*	Q	Major pigments are chlorophyll a and b
C	*Fucus*	R	Brown algae with haplo-diplontic life cycle
D	*Polysiphonia*	S	Lack flagella

Codes:

	A	**B**	**C**	**D**
1.	P	S	Q	R
2.	Q	P	R	S
3.	Q	R	P	S
4.	R	Q	S	P

120. In which of the following, the notochord is present in the embryonic stage?

1.	All chordates	2.	Some chordates
3.	Only Vertebrates	4.	Non-chordates

121. All the following will be true for heterotrophic bacteria except:

1.	They are most abundant bacteria in nature.
2.	The majority are important decomposers.
3.	None of them is capable of fixing atmospheric nitrogen.
4.	Some are pathogens causing damage to human beings, crops, farm animals and pets.

122. Identify the incorrect statement:

1.	A cnidoblast is seen in Cnidaria
2.	Bioluminescence is well developed in some Ctenophores
3.	Protonephridia are excretory and osmoregulatory structures in flatworms
4.	Nematodes have a true coelom

123. Animals in the phylum Porifera are characterised by:

I:	Nematocysts for defence
II:	Radial symmetry
III:	Filter-feeding and porous bodies
IV:	Cellular grade of organisation

1. Only **I** and **II**
2. Only **III** and **IV**
3. Only **IV**
4. Only **II** and **III**

124. Match each item in **Column-I** with one in **Column-II** and select the correct match from the codes given:

	Column-I [Protist]		**Column-II [Feature of wall]**
A.	Diatoms	P.	Cell wall has stiff cellulose plates on outer surface
B.	Dinoflagellates	Q.	Cell wall embedded with silica
C.	Euglenoids	R.	Protein rich layer instead of a cell wall
D.	Slime moulds	S.	True walls in spores

Codes:

	A	B	C	D
1.	Q	P	R	S
2.	Q	P	S	R
3.	P	Q	R	S
4.	P	Q	S	R

125. Which structure in the cockroach's alimentary canal is responsible for grinding food particles?
1. Crop
2. Gizzard
3. Oesophagus
4. Rectum

126. Scientific name does not ensure

1.	Each organism has only one name
2.	Description of any organism lead to the same name of organism in any part of the world
3.	No two organisms have the same name
4.	Status of threat of extinction of that organism holding a specific scientific name.

127. The type of tissue shown in the figure is found in:

I:	Tendon
II:	Ligament
III:	Skin

1. Only **I** and **II**
2. Only **III**
3. Only **II**
4. **I**, **II** and **III**

128. Given below are two statements:

Statement I:	Ligaments are dense irregular tissue.
Statement II:	Cartilage is dense regular tissue.

In the light of the above statements, choose the correct answer from the options given below:

1.	**Statement I** is false but **Statement II** is true.
2.	Both **Statement I** and **Statement II** are true.
3.	Both **Statement I** and **Statement II** are false.
4.	**Statement I** is true but **Statement II** is false.

129. You discover a unicellular organism. You shall place this organism in Kingdom Monera and not in Kingdom Protista if the cell:

I:	has non-cellulosic cell wall composed of polysaccharide and amino acids
II:	has a nuclear membrane but it is not double layered

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

130. indica is-

1.	Specific epithet of mango
2.	Specific epithet of Indus valley species
3.	Specific epithet of wheat (indian species)
4.	Specific epithet of Rice (indian species)

131. The institute which encourages publication of local flora in India is
1. NBRI
2. FRI
3. BSI
4.IARI

132. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in:

1.	Bronchioles and fallopian tubes
2.	Bile duct and bronchioles
3.	Fallopian tubes and pancreatic duct
4.	Eustachian tube and salivary duct

133. Ciliated epithelium in humans is primarily found in:

1.	The walls of capillaries and alveoli in the lungs
2.	The lining of sweat glands and sebaceous glands
3.	The outermost layer of the skin (epidermis)
4.	The inner lining of the fallopian tubes and respiratory tract

134. Which one of the following sets of items in options 1 – 4 are correctly categorized with one exception in it?

	ITEMS	CATEGORY	EXCEPTION
1.	Kangaroo, Koala, wombat	Australian marsupials	Wombat
2.	Plasmodium, *Cuscuta*, *Trypanosoma*	Protozoan parasites	*Cuscuta*
3.	Typhoid, Pneumonia, Diphtheria	Bacterial diseases	Diphtheria
4.	UAA, UAG, UGA	Stop codons	UAG

135. Identify the correct statements:

Statement I:	Frog eye does not have closable eye lids.
Statement II:	A nictitating membrane provides protection to the eye, especially when the frog is swimming.

1. **Statement I** is correct; **Statement II** is incorrect
2. **Statement I** is incorrect; **Statement II** is incorrect
3. **Statement I** is correct; **Statement II** is correct
4. **Statement I** is incorrect; **Statement II** is correct

136. Select the option that does not represent the taxa at different levels:

1.	Chordate, Mammal, Dog
2.	Animal, Insect, Butterfly
3.	Plant, Monocot, Wheat
4.	Mammal, Insect, Dicot

137. The sporophyte of mosses is more elaborate than that in:

1.	Pteridophytes	2.	Gymnosperms
3.	Angiosperms	4.	Liverworts

138. The most common species of frog found in India is:

1.	*Rana temporaria*	2.	*Rana tigrina*
3.	*Bufo bufo*	4.	*Bufo melanostictus*

139. What is prerequisite for nomenclature?

1.	Description of organism
2.	Knowledge of which organism is considered for a particular name
3.	Identification
4.	All are correct

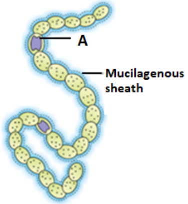
140. Which one of the following is considered important in the development of seed habit?

1.	Dependent sporophyte	2.	Heterospory
3.	Halplontic life cycle	4.	Free-living gametophyte

141. In the case of Poriferans, the spongocoel is lined with flagellated cells called:
1. Oscula
2. Choanocytes
3. Mesenchymal cells
4. Ostia

142. Numerical taxonomy, an approach now easily carried out using computers, is based on:
1. Chemical constituents of plants
2. Chromosome number and behaviour
3. External morphological features only
4. All observable characteristics

143. In the given diagram showing filamentous *Nostoc*, the structure marked as A:



1.	is a resting spore
2.	is involved in nitrogen fixation
3.	is gas vacuole
4.	is where carbon dioxide concentration mechanism is applied

144. Match List-I with List-II:

	List-I		List-II
A.	Squamous Epithelium	I.	Goblet cells of alimentary canal
B.	Ciliated Epithelium	II.	Inner lining of pancreatic ducts
C.	Glandular Epithelium	III.	Walls of blood vessels
D.	Compound Epithelium	IV.	Inner surface of Fallopian tubes

Choose the correct answer from the options given below:
1. A-II,B-III,C-I,D-IV
2. A-II,B-IV,C-III,D-I
3. A-III,B-I,C-II,D-IV
4. A-III,B-IV,C-I,D-II

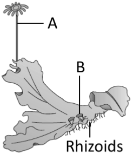
145. The cloaca in a frog is a small, median chamber that is used to pass:

1.	only faecal matter to the exterior.
2.	only faecal matter and urine but not sperms to the exterior.
3.	faecal matter, urine and sperms to the exterior.
4.	faecal matter and sperms but not urine to the exterior.

146. Simple squamous epithelium is found in areas like the alveoli of the lungs and blood vessels. Which of the following best explains why this tissue is suited for diffusion?

1.	It is composed of multiple layers of cells, providing strength.
2.	It has cilia that move substances across the surface.
3.	It is thin and flat, forming a single layer for efficient exchange.
4.	It secretes mucus to trap particles and prevent damage.

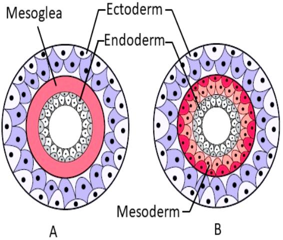
147. Study the given figure and select the correct statement/s:



A:	It is the male thallus of Marchantia.
B:	A is antheridiophore and B are gemma cups.

1. Only **A**
2. Only **B**
3. Both **A** and **B**
4. Neither **A** nor **B**

148. The diagram given below shows the germs layer. The animals having structures shown in the figures A and B are respectively called



1.	Diploblastic, Triploblastic	2.	Triploblastic, Diploblastic
3.	Diploblastic, Diploblastic	4.	Triploblastic, Triploblastic

149. The fungal class ascomycetes does not include:
1. *Morels and truffles*
2. *Neurospora and Claviceps*
3. *Aspergillus, Saccharomyces*
4. Puff balls and smuts

150. Which of the following statement is incorrect?

1.	Taxa can indicate categories at very different levels.
2.	systematics takes into account evolutionary relationship between organisms.
3.	Modern taxonomy do not take into account ecological information about organisms.
4.	Each rank or taxon represents unit of classification.

151. Match each item in Column I with one item in Column II and select the best match from the codes given:

	Column I		**Column II**
A.	Cyanobacteria	P.	Asexual reproduction
B.	Archaebacteria	Q.	Silica cell wall
C.	Diatoms	R.	Methanogens
D.	Deuteromycetes	S.	Nitrogen fixation

Codes:

	A	B	C	D
1.	S	R	Q	P
2.	Q	S	R	P
3.	P	Q	S	R
4.	R	S	P	Q

152. Male frogs can be distinguished from the female frogs by the presence of

A:	A sound-producing vocal sac
B:	A copulatory pad on the first digit of the forelimbs

1. Only **A** is correct
2. Only **B** is correct
3. Both **A** and **B** are correct
4. Both **A** and **B** are incorrect

153. Biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella are involved in:

1.	Asexual reproduction in brown algae
2.	Asexual reproduction in red algae
3.	Sexual reproduction in brown algae
4.	Sexual reproduction in red algae

154. Systematics does not include-
1. Identification
2. Nomenclature
3. Classification
4. Evolutionary relationships between environment and organisms

155. Consider the given two statements:

Statement I:	Coelenterates, ctenophores and echinoderms have bilateral symmetry as larvae and radial symmetry as adults.
Statement II:	Animals like annelids and arthropods exhibit bilateral symmetry.

1.	**Statement I** is correct; **Statement II** is incorrect
2.	**Statement I** is correct; **Statement II** is correct
3.	**Statement I** is incorrect; **Statement II** is incorrect
4.	**Statement I** is incorrect; **Statement II** is correct

156. Which of the following correctly differentiates cartilaginous fishes (Chondrichthyes) and bony fishes (Osteichthyes)?

Feature	Cartilaginous Fishes (Chondrichthyes)	Bony Fishes (Osteichthyes)
A	Skin has placoid scales	Skin has cycloid/ctenoid scales
B	Mouth is mostly ventral	Mouth is mostly terminal
C	Gill slits are separate and without an operculum	Gill slits are covered by an operculum
D	Air bladder is absent	Air bladder is present for buoyancy regulation

Choose the correct option:
1. A and B only
2. A and C only
3. A, C, and D only
4. A, B, C and D

157. Viroids are different from viruses because they consist only of:
1. DNA without a protein coat
2. Proteins without nucleic acid
3. RNA without a protein coat
4. Both DNA and RNA without a protein coat

158. By increasing the area of observation…..and ……… of organisms will increase.
1. Size and variety
2. Range and variety
3. Efficiency and range
4. Growth and variety

159. The members of deuteromycetes reproduce only by asexual spores called as:

1.	Zoospores	2.	Sporangiospores
3.	Aplanospores	4.	Conidia

160. Which of the following statements is true about zoospores and gametes in brown algae?

1.	Both zoospores and gametes in brown algae are motile and have equal sized flagella.
2.	Zoospores are non-motile while gametes are motile, both lacking flagella.
3.	Zoospores are motile with two flagella, while gametes are non-motile.
4.	Zoospores and gametes are both motile and similar in the number of flagella.

161. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I [Organ]		**Column II [Lining epithelium]**
A	PCT	P	Simple squamous
B	Small intestine	Q	Ciliated columnar
C	Alveoli	R	Columnar brush bordered
D	Fallopian tube	S	Cuboidal brush bordered

Codes:

	A	**B**	**C**	**D**
1.	R	S	P	Q
2.	R	S	Q	P
3.	S	R	P	Q
4.	S	R	Q	P

162. Specialized cells for fixing atmospheric nitrogen in *Nostoc* are:
1. Heterocysts
2. Hormogonia
3. Nodules
4. Akinetes

163. How many digits do the hind limbs of a frog have?
1. Three
2. Four
3. Five
4. Six

164. Sapindales is which taxon?

1.	Order	2.	Class
3.	Genus	4.	Species

165. The asexual spores of deuteromycetes are:

1.	Aplanospores	2.	Conidia
3.	Zoospores	4.	Basidiospores

166. How many pairs of spiracles are present in a cockroach?
1. 5 pairs
2. 8 pairs
3. 10 pairs
4. 12 pairs

167. Evolutionarily, the first terrestrial plants to possess vascular tissues – xylem and phloem, are:
1. Bryophytes
2. Pteridophytes
3. Gymnosperms
4. Angiosperms

168. Earlier classification systems included bacteria, blue green algae and fungi under 'plants'. The problem with this inclusion was:

I:	Bacteria and blue green algae have prokaryotic cell organization unlike other members included in 'plants'.
II:	Fungi and many bacteria are heterotrophic unlike autotrophic members included in 'plants'.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

169. Consider the given two statements:

Assertion(A):	Artificial systems of classification of organisms are simple and acceptable to most scientists.
Reason (R):	Artificial classification systems take into account the evolutionary relationships between living organisms.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A).**
3.	**(A)** is True; **(R)** is False
4.	Both **(A)** and **(R)** are False

170. Wheat comes under which family
1. Anacardiaceae
2. Poaceae
3. Muscidae
4. Poals

171. Consider the following two statements:
Nitrogen-fixing organisms belong to
I: Kingdom Monera
II: Kingdom Protista

1.	Only **I** is correct
2.	Only **II** is correct
3.	Both **I** and **II** are correct
4.	Neither **I** nor **II** are correct

172. On an average a female cockroach produces:
1. 9-10 oothecae , each containing 14-16 eggs
2. 14-16 oothecae , each containing 9-10 eggs
3. 5-6 oothecae , each containing 9-10 eggs
4. 9-10 oothecae , each containing 5-6 eggs

173.

Statement I:	The ground substance of bone is solid, pliable and resists compression.
Statement II:	The bone marrow of all bones is the site of production of blood cells.

In light of the above statements, choose the most appropriate answer from the options given below.
1. Both **Statement I** and **Statement II** are correct
2. Both **Statement I** and **Statement II** are incorrect.
3. **Statement I** is correct but **Statement II** is incorrect
4. **Statement I** is incorrect but **Statement II** is correct

174. Consider the given two statements:

Assertion (A):	The gastro-intestinal tract of *Rana tigrina* is short in length.
Reason (R):	*Rana tigrina* is a carnivore.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.
3.	**(A)** is True; **(R)** is False
4.	Both **(A)** and **(R)** are False

175. The terms for 'summer sleep' and 'winter sleep' seen in frogs respectively are:

1.	Aestivation and Hibernation
2.	Hibernation and Aestivation
3.	Diapause and Cryptobiosis
4.	Cryptobiosis and Diapause

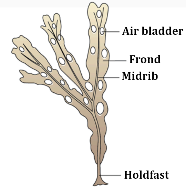
176. Which one of the following animals is correctly matched with its particular named taxonomic category?

1.	Cuttlefish	–	Mollusca, a class
2.	Humans	–	Primata, the family
3.	Housefly	–	*Musca*, an order
4.	Tiger	–	*tigris*, the species

177. Which protists are all heterotrophs and live as predators or parasites?

1.	Desmids	2.	Slime moulds
3.	Protozoans	4.	Euglenoids

178. The alga shown in the given figure is most likely:



1.	*Chara*	2.	*Spirogyra*
3.	*Fucus*	4.	Gelidium

179. Polymoniales includes-
1. Convolvulaceae
2. Solanaceae
3. Both 1 and 2
4. Liliaceae only

180. The pathogen responsible for African Sleeping Sickness belongs to which group of protozoa?

1.	Ciliates	2.	Amoeboid
3.	Flagellates	4.	Sporozoans

Mitoprep

Sapno Ki Udaan